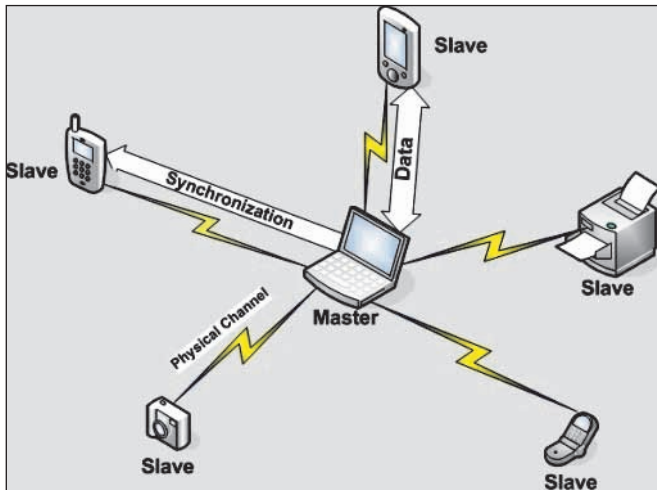


technique in Bluetooth devices. If a device is on standby, it waits for active connections. Hence, it “listens” every 1.28 seconds, for messages from other devices. Again, as mentioned before, Bluetooth prefers to access a range of channels, and the listening takes place across 32 hop frequencies for that specific type of unit. As mentioned earlier, the Bluetooth Specification has exact listings for each “profile”, hence mobile phones will search across a different range of frequencies compared to other devices.

The concept of a master-slave is set up once one unit senses another Bluetooth device in the region. The first device that found the other unit assumes the role of the master unit in the development of the piconet. Of course, since all Bluetooth radios are identical, any unit may assume the role of the master or the slave. The only cause for selection is which detects the other first.

When two Bluetooth devices establish a connection, they have created a type of personal area network called a piconet. Each piconet can contain up to eight different Bluetooth devices. Within each piconet, one device serves as the master, while the other seven devices function as slaves. Any individual device can



Piconet